



2014 Half-Yearly Examination

FORM V

MATHEMATICS 3 UNIT

Monday 19th May 2014

General Instructions

- Writing time — 1 hour 30 minutes
- Write using black or blue pen.
- Board-approved calculators and templates may be used.

Total — 80 Marks

- All questions may be attempted.

Section I – 8 Marks

- Questions 1–8 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II – 72 Marks

- Questions 9–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your name, class and master on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Nine.
- Write your name and master on this question paper and submit it with your answers.

5A: BDD

5E: PKH

5B: MLS

5F: BR

5C: LYL

5G: SG

5D: LRP

Checklist

- SGS booklets — 6 per boy
- Multiple choice answer sheet
- Candidature — 125 boys

Examiner

BDD

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

Given $f(x) = \frac{1}{3x}$, what is the value of $f'(2)$?

- | | |
|---------------------|--------------------|
| (A) $-\frac{1}{12}$ | (B) $-\frac{1}{6}$ |
| (C) $\frac{1}{3}$ | (D) $-\frac{3}{4}$ |

QUESTION TWO

Which of the following statements is NOT correct?

- | | |
|---|---|
| (A) $1 + \tan^2 \theta = \sec^2 \theta$ | (B) $\cos(90^\circ - \theta) = \sin \theta$ |
| (C) $\tan \theta = \frac{\sin \theta}{\cos \theta}$ | (D) $\sec \theta = \frac{1}{\sin \theta}$ |

QUESTION THREE

The recurring decimal $0.002020202\dots$ may be regarded as the limiting sum of a GP with common ratio:

- | | |
|-----------|-----------|
| (A) 0.001 | (B) 0.002 |
| (C) 0.01 | (D) 0.02 |

QUESTION FOUR

Which of the following is a correct limit statement about $y = \frac{1}{x-2}$?

- | | |
|--|---|
| (A) $y \rightarrow \infty$ as $x \rightarrow 2^-$. | (B) $y \rightarrow \infty$ as $x \rightarrow -2$. |
| (C) $y \rightarrow -\infty$ as $x \rightarrow 2^-$. | (D) $y \rightarrow -\infty$ as $x \rightarrow -2$. |

QUESTION FIVE

The expression $\frac{\log_3 x^2}{\log_3 x}$ simplifies to:

- | | |
|-----------------------|----------------|
| (A) x | (B) $\log_3 x$ |
| (C) $\log_3(x^2 - x)$ | (D) 2 |

Exam continues next page ...

QUESTION SIX

A line passing through the point of intersection of $2x + 3y - 1 = 0$ and $4x - 9y + 3 = 0$ may be written in the form $2x + 3y - 1 + k(4x - 9y + 3) = 0$ for some real number k .

For what value of k does this line pass through the point $(2, 1)$?

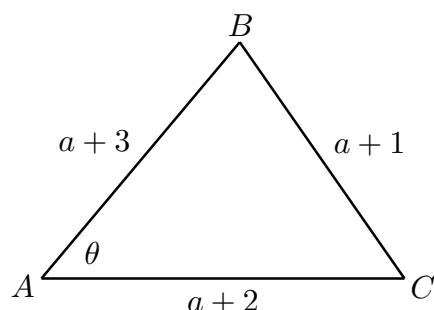
- (A) $k = 2$ (B) $k = -2$
(C) $k = -3$ (D) $k = 3$

QUESTION SEVEN

The common ratio of the geometric sequence $\sqrt{2} - 1, 6\sqrt{2} - 3, 45\sqrt{2} + 9, \dots$ is:

- (A) $3 + \sqrt{2}$ (B) $9 + 3\sqrt{2}$
(C) $5\sqrt{2} - 2$ (D) $\frac{3 - \sqrt{2}}{9}$

QUESTION EIGHT



The value of $\cos \theta$ in the triangle above, for $a > -1$, is:

- (A) $\frac{a + 2}{a + 3}$ (B) $\frac{a + 6}{2a + 6}$
(C) $\frac{a^2 + 12a + 14}{2(a + 3)(a + 2)}$ (D) $\frac{a^2 + 12}{2a^2 + 10a + 12}$

_____ End of Section I _____

Exam continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION NINE (12 marks) Use a separate writing booklet. **Marks**

(a) Simplify $\frac{x^2 - 3x - 10}{x + 2}$. 1

(b) Write down the natural domain of the function $f(x) = \sqrt{x^2 - 4}$. 1

(c) Differentiate:

(i) $y = 3x^2 + 2x + 4$ 1

(ii) $y = 6\sqrt{x}$ 1

(d) Find the sum of the first twelve terms of the sequence 3, 7, 11, ... 2

(e) Given the geometric sequence 1536, 768, 384, ...

(i) find the common ratio, 1

(ii) find the tenth term. 1

(f) Determine whether the function $f(x) = 2^x - 2^{-x}$ is even, odd or neither. 1

(g) Consider the function $f(x) = 5x^2 + 4x$.

(i) Simplify the expression $f(x + h) - f(x)$. 2

(ii) Hence use the formula 1

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

to differentiate $f(x) = 5x^2 + 4x$ from first principles.

QUESTION TEN (12 marks) Use a separate writing booklet.

Marks

- (a) Evaluate $\sum_{k=2}^4 k^2$. **1**
- (b) Let A and B be points with coordinates $A(-9, 7)$ and $B(6, 3)$. Find the coordinates of the point $P(x, y)$ that divides AB in the ratio $4 : 1$. **2**
- (c) Suppose T divides FG externally in the ratio $3 : 5$. In what ratio does F divide TG ? **1**
- (d) Solve:
- (i) $2\sin^2 \theta - \sin \theta - 1 = 0$, for $0^\circ \leq \theta \leq 360^\circ$. **3**
- (ii) $\cos 2\theta = \frac{\sqrt{3}}{2}$, for $0^\circ \leq \theta \leq 360^\circ$. **3**
- (e) Consider the limiting sum $45 + 15 + 5 + \dots$.
- (i) Give a reason why the limiting sum is known to exist. **1**
- (ii) Find the limiting sum. **1**

QUESTION ELEVEN (12 marks) Use a separate writing booklet.

Marks

- (a) Find the equation of the tangent to $y = x^2 - 3x + 2$ at $x = 4$. **3**
- (b) Differentiate:
- (i) $y = (2x - 1)(x + 3)$ **2**
- (ii) $y = (7 - 4x)^9$ **2**
- (c) Consider the graphs of $y = x^2$ and $y = \frac{1}{2}x + 3$.
- (i) Determine the two points of intersection of the graphs. **2**
- (ii) Shade the region where $y > x^2$ and $y \leq \frac{1}{2}x + 3$. **3**
- Be careful to mark the corners and boundaries of the region correctly.

QUESTION TWELVE (12 marks) Use a separate writing booklet.

Marks

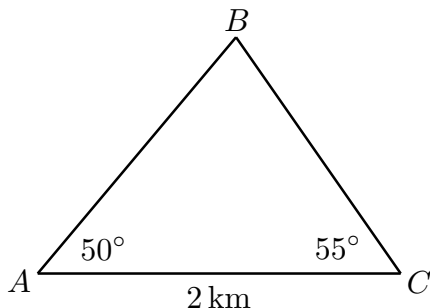
(a) Solve the inequation $\frac{2x}{x-3} > 4$.

3

(b) A geometric sequence has fifth term 18 and ninth term 1458. Find the common ratio and the first term.

3

(c)



From two points A and C on a straight horizontal road a balloon at B directly above the road is observed to have angles of elevation of 50° and 55° respectively.

(i) Show that $BC \doteq 1.59$ km.

2

(ii) Find the height of the balloon above the road. Give your answer correct to 3 significant figures.

1

(d) The line ℓ has equation $12x - 5y + c = 0$ for some constant c .

(i) Write down a simplified expression for the perpendicular distance from ℓ to a point $P(x_0, y_0)$.

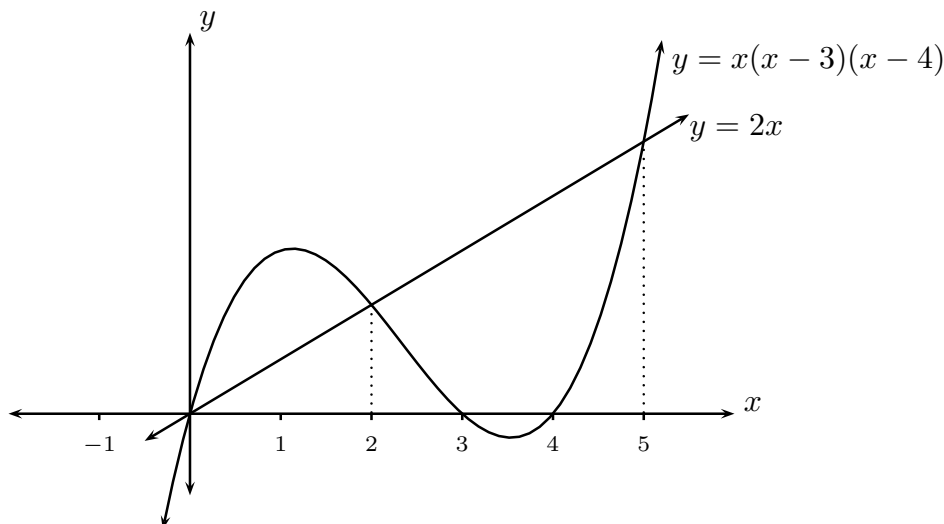
1

(ii) The line is known to be a tangent to the circle $(x - 2)^2 + (y - 3)^2 = 1$. Find the possible values of c .

2

QUESTION THIRTEEN (12 marks) Use a separate writing booklet.**Marks**

(a)

Use the graph above to solve the inequation $x(x - 3)(x - 4) > 2x$.**1**(b) Prove that $\frac{1}{1 - \cos \theta} + \frac{1}{1 + \cos \theta} = 2 \operatorname{cosec}^2 \theta$.**2**(c) Use mathematical induction to prove that for all positive integers n ,**3**

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \cdots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}.$$

(d) Consider the graph of $y = \frac{(x-4)^2}{4x(x-2)}$.(i) Find any intercepts with the x and y axes.**1**

(ii) Write down the equation(s) of any vertical asymptote(s).

1

(iii) Find the equation of the horizontal asymptote.

1

(iv) Copy and fill in the following table:

1

x	-1	1	3
y			

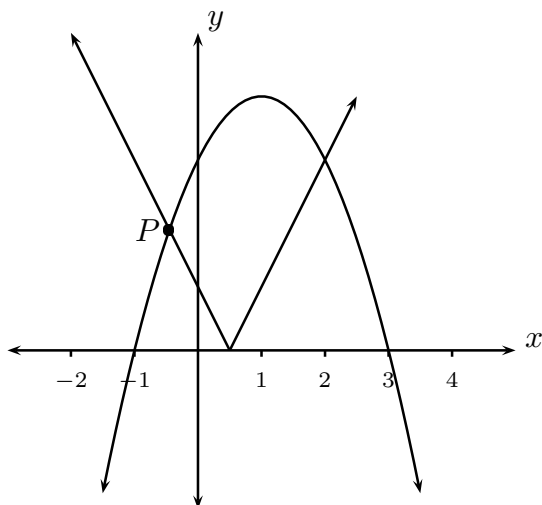
(v) Use parts (i)-(iv) to sketch the curve.

2

QUESTION FOURTEEN (12 marks) Use a separate writing booklet.**Marks**

- (a) For what values of x is the tangent to $y = \left(\frac{8}{x} - x^2\right)^5$ horizontal? 2

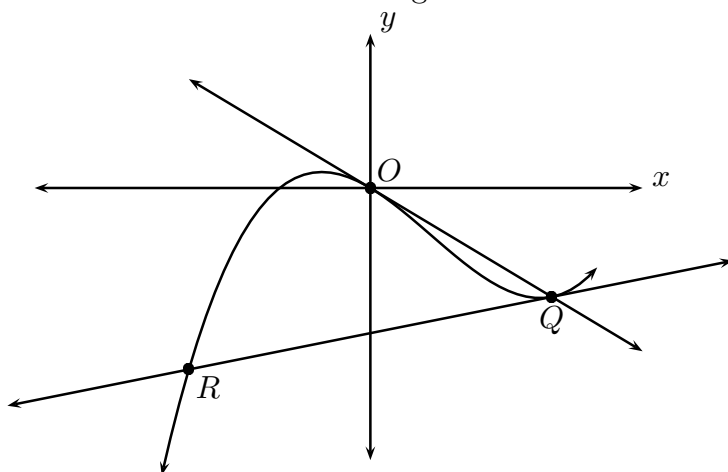
(b)



The diagram above shows the graphs of $y = 3x + 3 - x^2$ and $y = |2x - 1|$. 3

Find the exact value of the x -coordinate of the point P where the graphs intersect in the second quadrant.

- (c) Consider a cubic $y = ax^3 + bx^2 + cx$, where $a, b \neq 0$, passing through the origin O . One such cubic is shown in the diagram below.



- (i) Find the equation of the tangent at O for $y = ax^3 + bx^2 + cx$. 1
- (ii) Find the coordinates of the point Q where the tangent in part (i) meets the cubic again. 2
- (iii) Show that the y -intercept of the tangent at Q is independent of c . 2
- (iv) Suppose the tangent at Q meets the cubic again at R . Show that the x -coordinates of R , O and Q form an arithmetic sequence. 2

————— End of Section II —————

END OF EXAMINATION



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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

NAME:

CLASS: MASTER:

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

SECTION I - Multiple Choice**QUESTION ONE**

$$\begin{aligned}f(x) &= \frac{1}{3x} \\&= \frac{1}{3}x^{-1}\end{aligned}$$

$$\begin{aligned}\text{Hence } f'(x) &= \frac{1}{3} \times (-1)x^{-2} \\&= -\frac{1}{3x^2}\end{aligned}$$

$$\begin{aligned}f'(2) &= -\frac{1}{3 \times (2)^2} \\&= -\frac{1}{12}\end{aligned}$$

The correct answer is A.

QUESTION TWO

$\sec \theta = \frac{1}{\cos \theta}$, hence the incorrect statement is D.

QUESTION THREE

$$\begin{aligned}0.002020202\dots &= 0.002 + 0.00002 + 0.0000002 + \dots + 0.000000002 + \dots \\&= 0.002 + 0.002 \div 100 + 0.002 \div 10000 + 0.002 \div 1000000 + \dots\end{aligned}$$

Each successive term is divided by 100, hence the common ratio is $\frac{1}{100} = 0.01$.

The correct answer is C.

QUESTION FOUR

The function has an asymptote at $x = 2$, so we want a limit statement for $x \rightarrow 2$, that is A or C.

If x approaches 2 from below, it is less than 2, so $x - 2$ is negative.

Hence the correct answer is C.

QUESTION FIVE

$$\begin{aligned}\frac{\log_3 x^2}{\log_3 x} &= \frac{2 \times \log_3 x}{\log_3 x} \\&= 2\end{aligned}$$

Hence the correct answer is D.

QUESTION SIX

If we substitute the point $(2, 1)$ into the expression $2x + 3y - 1 + k(4x - 9y + 3) = 0$, then we get:

$$2(2) + 3(1) - 1 + k(4(2) - 9(1) + 3) = 0$$

$$6 + 2k = 0$$

$$k = -3$$

Hence the correct answer is C.

QUESTION SEVEN

$$\begin{aligned} r &= \frac{6\sqrt{2} - 3}{\sqrt{2} - 1} \\ &= \frac{(6\sqrt{2} - 3)(\sqrt{2} + 1)}{\sqrt{2}^2 - 1^2} \\ &= \frac{12 + 6\sqrt{2} - 3\sqrt{2} - 3}{1} \\ &= 9 + 3\sqrt{2} \end{aligned}$$

Hence B.

QUESTION EIGHT

By the cosine rule,

$$\begin{aligned} \cos \theta &= \frac{AB^2 + AC^2 - BC^2}{2 \times AB \times AC} \\ &= \frac{(a+3)^2 + (a+2)^2 - (a+1)^2}{2(a+3)(a+2)} \\ &= \frac{(a+2)^2 + (2a+4)(2)}{2(a+3)(a+2)} \quad \text{(difference of squares)} \\ &= \frac{(a+2)((a+2)+4)}{2(a+3)(a+2)} \\ &= \frac{(a+6)}{2(a+3)} \end{aligned}$$

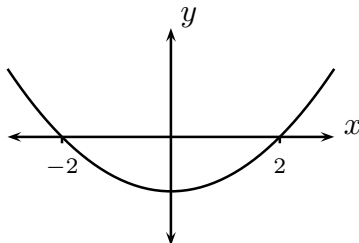
Hence the correct answer is B.

(The pupil could expand the numerator as an alternative to using difference of squares.)

SECTION II - Written Response**QUESTION NINE**

$$(a) \frac{x^2 - 3x - 10}{x + 2} = \frac{(x - 5)(x + 2)}{(x + 2)} \\ = x - 5$$

(b) The function is defined where $x^2 - 4 \geq 0$.



Thus the domain is $x \leq -2$ or $x \geq 2$.

$$(c) (i) y' = 6x + 2$$

$$(ii) y = 6\sqrt{x} \\ = 6x^{\frac{1}{2}}$$

$$y' = 6 \times \left(\frac{1}{2}\right)x^{-\frac{1}{2}}$$

$$y' = \frac{3}{\sqrt{x}}$$

(d) This is an arithmetic progression. The first term is $a = 3$ and the common difference is 4.

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

$$S_{12} = \frac{12}{2}(2(3) + (12 - 1)(4))$$

$$S_{12} = 6 \times 50$$

$$S_{12} = 300$$

$$(e) (i) r = 768 \div 1536$$

$$= \frac{1}{2}$$

$$(ii) T_n = ar^{n-1}$$

$$T_{10} = 1536 \times \left(\frac{1}{2}\right)^9$$

$$= 1536 \div 512$$

$$= 3$$

$$(f) f(x) = 2^x - 2^{-x}$$

$$f(-x) = 2^{(-x)} - 2^{-(-x)}$$

$$f(-x) = 2^{-x} - 2^x$$

$$f(-x) = -(2^x - 2^{-x})$$

$$f(-x) = -f(x)$$

hence the function has odd symmetry.

$$\begin{aligned} \text{(g) (i) } f(x+h) - f(x) &= 5(x+h)^2 + 4(x+h) - (5x^2 + 4x) \\ &= 5(x^2 + 2xh + h^2) + 4x + 4h - 5x^2 - 4x \\ &= 10xh + 5h^2 + 4h \end{aligned}$$

$$\begin{aligned} \text{(ii) } f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{10xh + 5h^2 + 4h}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(10x + 4 + 5h)}{h} \\ &= \lim_{h \rightarrow 0} (10x + 4 + 5h) \\ &= 10x + 4 \end{aligned}$$

QUESTION TEN

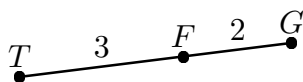
$$\begin{aligned} \text{(a) } \sum_{k=2}^4 k^2 &= 2^2 + 3^2 + 4^2 \\ &= 4 + 9 + 16 \\ &= 29 \end{aligned}$$

$$\begin{aligned} \text{(b) } x &= \frac{kx_2 + \ell x_1}{k + \ell} \\ &= \frac{4(6) + 1(-9)}{4 + 1} \\ &= 3 \end{aligned}$$

$$\begin{aligned} y &= \frac{ky_2 + \ell y_1}{k + \ell} \\ &= \frac{4(3) + 1(7)}{4 + 1} \\ &= \frac{19}{5} \end{aligned}$$

The required point is $P(3, \frac{19}{5})$.

(c)



$$FT : TG = -3 : 5 \text{ so } TF : FG = 3 : 2.$$

$$\begin{aligned} \text{(d) (i) } 2 \sin^2 \theta - \sin \theta - 1 &= 0 \\ (2 \sin \theta + 1)(\sin \theta - 1) &= 0 \end{aligned}$$

$$\text{Hence } 2 \sin \theta + 1 = 0$$

$$\sin \theta = -\frac{1}{2}$$

$$\theta = 210^\circ, 330^\circ$$

$$\text{Or } \sin \theta - 1 = 0$$

$$\sin \theta = 1$$

$$\theta = 90^\circ$$

(ii) Let $u = 2\theta$. Then

$$\cos 2\theta = \frac{\sqrt{3}}{2}, \quad \text{for } 0^\circ \leq \theta \leq 360^\circ$$

$$\cos u = \frac{\sqrt{3}}{2}, \quad \text{for } 0^\circ \leq u \leq 720^\circ$$

$$u = 30^\circ, 330^\circ, 390^\circ, 690^\circ$$

$$2\theta = 30^\circ, 330^\circ, 390^\circ, 690^\circ$$

$$\theta = 15^\circ, 165^\circ, 195^\circ, 345^\circ$$

(e) This is a GP with common ratio $r = \frac{1}{3}$.

(i) Since $-1 < r < 1$, the limiting sum exists.

$$\begin{aligned} \text{(ii) } S_\infty &= \frac{a}{1-r} \\ &= \frac{45}{1-\frac{1}{3}} \times 3 \\ &= \frac{135}{3-1} \\ &= 67.5 \end{aligned}$$

QUESTION ELEVEN

$$\begin{aligned} \text{(a) } y' &= 2x - 3 \\ &= 2(4) - 3 \quad \text{at } x = 4 \\ &= 5 \end{aligned}$$

$$\begin{aligned} \text{and } y &= x^2 - 3x + 2 \\ &= (4)^2 - 3(4) + 2 \\ &= 6 \end{aligned}$$

$$\text{Hence } (y - y_1) = m(x - x_1)$$

$$(y - 6) = 5(x - 4)$$

$$y = 5x - 14$$

$$\text{(b) (i) } y = (2x - 1)(x + 3)$$

$$= 2x^2 + 5x - 3$$

$$y' = 4x + 5$$

(ii) Let $y = 7 - 4x$.

$$\text{Then } y = (7 - 4x)^9$$

$$= u^9$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$= 9u^8 \times -4$$

$$= -36(7 - 4x)^8$$

.....

$$\frac{dy}{du} = 9u^8$$

$$\frac{du}{dx} = -4$$

(c) The graphs intersect when:

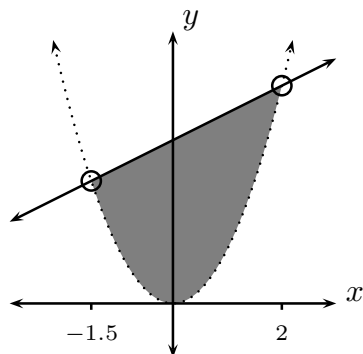
$$x^2 = \frac{1}{2}x + 3$$

$$2x^2 - x - 6 = 0$$

$$(2x + 3)(x - 2) = 0$$

Hence they intersect at $(-\frac{3}{2}, \frac{9}{4})$ and $(2, 4)$.

(i)



QUESTION TWELVE

(a) $\frac{2x}{x-3} > 4, \quad x \neq 3$

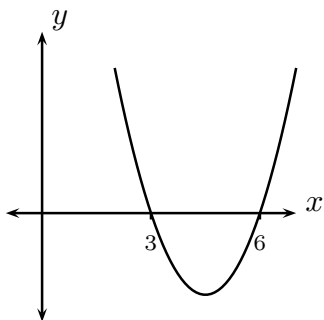
$$\boxed{\times (x-3)^2}$$

$$2x(x-3) > 4(x-3)^2$$

$$0 > 4(x-3)^2 - 2x(x-3)$$

$$0 > (x-3)(4x-12-2x)$$

$$0 > (x-3)(2x-12)$$



Hence $3 < x < 6$.

(b) We have $T_5 = ar^4$ and $T_9 = ar^8$. Hence

$$ar^4 = 18 \quad \boxed{1}$$

$$ar^8 = 1458 \quad \boxed{2}$$

Thus $\boxed{2} \div \boxed{1}$ gives:

$$r^4 = 81$$

$$r = \pm 3$$

Hence from $\boxed{1}$,

$$a = \frac{18}{81}$$

$$= \frac{2}{9}$$

- (c) Note first that $\angle ABC = 75^\circ$ (angle sum of triangle). Now by the Sine Rule;

$$\frac{BC}{\sin 50^\circ} = \frac{2}{\sin 75^\circ}$$

$$BC = \frac{2 \times \sin 50^\circ}{\sin 75^\circ}$$

$$\doteq 1.59 \text{ km}$$

Construct $BX \perp AC$ with X on AC . Thus BX is the required height of the balloon above the road. Then by right-angled triangle trigonometry,

(i) $\frac{BX}{1.59} = \sin 55^\circ$

$$BX \doteq 1.59 \times \sin 55^\circ$$

$$\doteq 1.30 \text{ km}$$

- (d) (i) The perpendicular distance d from ℓ to $P(x_0, y_0)$ is given by

$$d = \frac{|12x_0 - 5y_0 + c|}{\sqrt{12^2 + 5^2}}$$

$$= \frac{|12x_0 - 5y_0 + c|}{13}$$

- (ii) If ℓ is a tangent, then its distance d from the centre of the circle is exactly the radius 1.

Thus

$$\frac{|12(2) - 5(3) + c|}{13} = 1$$

$$|12(2) - 5(3) + c| = 13$$

$$|c + 9| = 13$$

Hence $c + 9 = 13$ or $c + 9 = -13$.

Thus $c = 4$ or $c = -22$.

QUESTION THIRTEEN

- (a) We want the x values for which the cubic $y = x(x - 3)(x - 4)$ is ABOVE the line $y = 2x$.

Thus the solution set is $0 < x < 2$ or $x > 5$.

(b) $\text{LHS} = \frac{1}{1 - \cos \theta} + \frac{1}{1 + \cos \theta}$

$$= \frac{(1 + \cos \theta) + (1 - \cos \theta)}{(1 + \cos \theta)(1 - \cos \theta)}$$

$$= \frac{2}{1 - \cos^2 \theta}$$

$$= \frac{2}{\sin^2 \theta}$$

$$= 2 \operatorname{cosec}^2 \theta$$

- (c)

A. First we shall check the case $n = 1$;

$$\text{LHS} = \frac{1}{1 \times 3} \quad \text{RHS} = \frac{1}{2 \times 1 + 1}$$

$$= \frac{1}{3} \quad = \frac{1}{3}$$

Thus the result is true for $n = 1$.

B. Now we assume the result holds for $n = k$, that is that

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \cdots + \frac{1}{(2k-1)(2k+1)} = \frac{k}{2k+1}.$$

We want to show that the result holds for $n = k + 1$, namely that

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \cdots + \frac{1}{(2(k+1)-1)(2(k+1)+1)} = \frac{k+1}{2(k+1)+1}.$$

Consider the LHS of this expression;

$$\begin{aligned} LHS &= \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \cdots + \frac{1}{(2(k+1)-1)(2(k+1)+1)} \\ &= \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \cdots + \frac{1}{(2k-1)(2k+1)} + \frac{1}{(2(k+1)-1)(2(k+1)+1)} \\ &= \frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)} \\ &= \frac{k(2k+3)+1}{(2k+1)(2k+3)} \\ &= \frac{2k^2+3k+1}{(2k+1)(2k+3)} \\ &= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)} \\ &= \frac{(k+1)}{(2k+3)} \\ &= RHS \end{aligned}$$

C. The result now follows for all positive integers $n = 1, 2, \dots$, by the principle of mathematical induction.

(d) (i) There is no y -intercept, since $x = 0$ is not on the domain.
The x -intercept is $(4, 0)$.

(ii) The vertical asymptotes are $x = 0$ and $x = 2$.

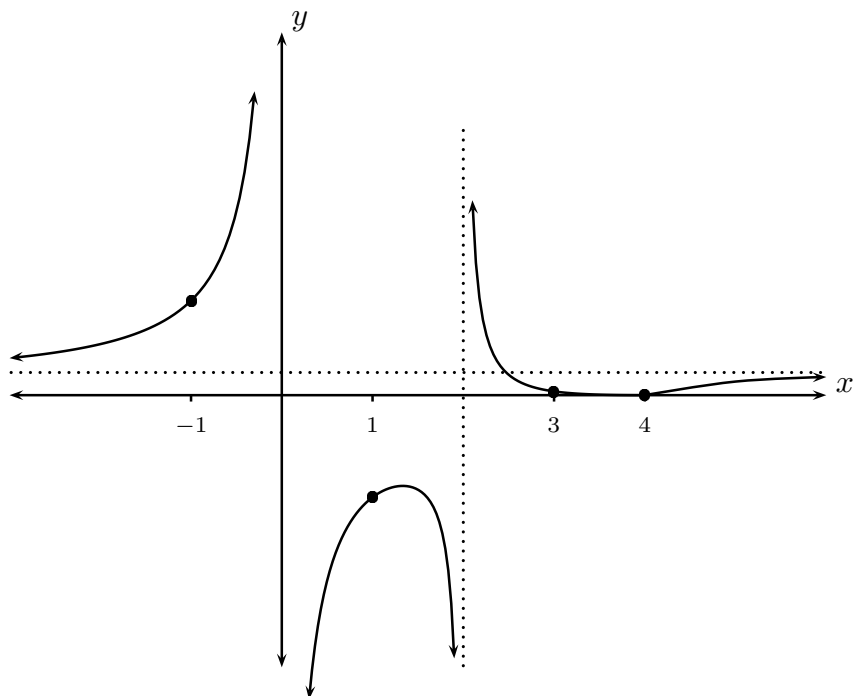
$$\begin{aligned} \text{(iii) } y &= \frac{(x-4)^2}{4x(x-2)} \\ &= \frac{x^2 - 8x + 16}{4x^2 - 8x} \div x^2 \\ &= \frac{1 - \frac{8}{x} + \frac{16}{x^2}}{4 - \frac{8}{x}} \\ &\rightarrow \frac{1}{4} \quad \text{as } \frac{1}{x} \rightarrow 0 \end{aligned}$$

Hence the horizontal asymptote is $y = \frac{1}{4}$.

(iv)

x	-1	1	3
y	$\frac{25}{12}$	$\frac{9}{-4}$	$\frac{1}{12}$

(v)



QUESTION FOURTEEN

(a) Let $u = 8x^{-1} - x^2$, and then

Then $y = u^5$

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\ &= 5u^4 \times (-8x^{-2} - 2x) \\ &= 5\left(\frac{8}{x} - x^2\right)^4 \left(-\frac{8}{x^2} - 2x\right)\end{aligned}$$

$$\begin{aligned}\frac{dy}{du} &= 5u^4 \\ \frac{du}{dx} &= -8x^{-2} - 2x\end{aligned}$$

The tangent is horizontal when

$$\frac{8}{x} = x^2$$

$$8 = x^3$$

$$x = 2$$

$$\begin{aligned}\text{OR} \quad -\frac{8}{x^2} &= 2x \\ -4 &= x^3 \\ x &= -\sqrt[3]{4}\end{aligned}$$

(b) The graphs intersect where:

$$3x + 3 - x^2 = -(2x - 1)$$

$$0 = x^2 - 5x - 2$$

The solutions of this quadratic are:

$$\begin{aligned}x &= \frac{5 \pm \sqrt{25 - 4 \times 1 \times (-2)}}{2 \times 1} \\ &= \frac{5 \pm \sqrt{33}}{2}\end{aligned}$$

We need the negative root, i.e. $x = \frac{5 - \sqrt{33}}{2}$.

$$(c) \quad (i) \quad y' = 3ax^2 + 2bx + c \\ = c \quad \text{at } x = 0$$

The tangent to the cubic at $(0, 0)$ is $y = cx$.

(ii) This tangent intersects the cubic when

$$ax^3 + bx^2 + cx = cx$$

$$ax^3 + bx^2 = 0$$

$$x^2(ax + b) = 0$$

Hence it intersects at $x = 0$ and again at $x = -\frac{b}{a}$, $y = -\frac{cb}{a}$.

(iii) We need to find the equation of the tangent at Q . Thus

$$y' = 3ax^2 + 2bx + c$$

$$= 3a\left(-\frac{b}{a}\right)^2 + 2b\left(-\frac{b}{a}\right) + c \quad \text{at } x = -\frac{b}{a}$$

$$= \frac{3b^2}{a} - \frac{2b^2}{a} + c$$

$$= \frac{b^2}{a} + c$$

and the tangent is

$$y - y_1 = m(x - x_1)$$

$$y + \frac{cb}{a} = \left(\frac{b^2}{a} + c\right)\left(x + \frac{b}{a}\right)$$

$$y + \frac{cb}{a} = \left(\frac{b^2}{a} + c\right)x + \frac{b^3}{a^2} + \frac{cb}{a}$$

$$y = \left(\frac{b^2}{a} + c\right)x + \frac{b^3}{a^2}$$

The y -intercept is $(0, \frac{b^3}{a^2})$, which is independent of c .

(iv) Now intersect THIS tangent and the original cubic:

$$ax^3 + bx^2 + cx = \left(\frac{b^2}{a} + c\right)x + \frac{b^3}{a^2}$$

$$ax^3 + bx^2 - \left(\frac{b^2}{a}\right)x - \frac{b^3}{a^2} = 0$$

$$ax^2\left(x + \frac{b}{a}\right) - \frac{b^2}{a}\left(x + \frac{b}{a}\right) = 0$$

$$\left(ax^2 - \frac{b^2}{a}\right)\left(x + \frac{b}{a}\right) = 0$$

$$a\left(x - \frac{b}{a}\right)\left(x + \frac{b}{a}\right)^2 = 0$$

The tangent and cubic intersect again at $x = \frac{b}{a}$.

The coordinates $\frac{b}{a}$, 0 and $-\frac{b}{a}$ form an AP with common difference $-\frac{b}{a}$.

BDD